Literature Status: Two Lebesgue-Type Questions for Quasi-Greedy Bases

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Status: literature already answered. Questions 1 and 2 in Section 3.4 of arXiv:1111.0460 are explicitly answered in the separate later paper arXiv:1207.0946. The first answer is sharpness by examples; the second is a universal affirmative inequality.

1 Original questions

The preliminary paper is Eugenio Hernández, *Lebesgue-Type inequalities for quasi-greedy bases* [1]. It defines the best N-term error σ_N, the expansional best error $\widetilde{\sigma}_N$, the greedy Lebesgue constant e_N, and the democracy ratio $\mu(N)$. Section 3.4, arXiv PDF page 7, asks the following.

Source questions. *Is the estimate $e_N \lesssim \mu(N) \log N$ sharp, in the sense that some quasi-greedy basis has $e_N \asymp \mu(N) \log N$? Is it always true that*

$$\widetilde{\sigma}_N(x) \lesssim \sigma_N(x) \log N$$

for a quasi-greedy basis?

The source labels these as Questions 1 and 2. Its following Question 3 is a different, broad characterization problem and is outside this status note.

2 Separate later answers

Garrigós, Hernández, and Oikhberg's separate paper with the same title, arXiv:1207.0946 [2], explicitly lists arXiv:1111.0460 as reference [10] and calls it the preliminary version written by the second author.

For Question 1, Theorem 1.1 on arXiv PDF page 2 proves the sharp formula

$$C_N \asymp \max\{\mu(N), k_N\}, \qquad k_N = \sup_{|A| \le N} \|S_A\|.$$

The remarks continuing on PDF page 3 observe that for a democratic quasi-greedy basis this becomes $C_N \asymp k_N \lesssim \log N$, and state that Section 6 constructs examples attaining the logarithmic bound, "answering a question posed in [10]." For these democratic examples $\mu(N) \asymp 1$, so $C_N \asymp \mu(N) \log N$. Thus Question 1 has an affirmative answer in general Banach spaces. The same remark notes a Hilbert-space distinction: there k_N cannot attain the full logarithmic order.

For Question 2, define

$$D_N = \sup_{x \neq 0} \frac{\widetilde{\sigma}_N(x)}{\sigma_N(x)}.$$

Theorem 1.2 on arXiv PDF page 3 proves, for every normalized basis,

$$\frac{k_N}{4} \leq D_N \leq 2k_N.$$

The immediately following remark combines the upper bound with the standard quasi-greedy estimate $k_N \lesssim \log N$ and explicitly says that this gives $\widetilde{\sigma}_N(x) \lesssim (\log N)\sigma_N(x)$, answering a question from [10].

For completeness, the upper estimate has a one-line projection explanation. If an N-term vector y is supported on A, then

$$x - S_A x = (I - S_A)(x - y),$$

and hence

$$\widetilde{\sigma}_N(x) \leq (1 + k_N)\sigma_N(x).$$

This already yields the requested logarithmic estimate for quasi-greedy bases; Theorem 1.2 sharpens the comparison to two-sided equivalence with k_N.

3 Scope and provenance

Both identifications are author-explicit, not new discoveries: the supporting paper names the preliminary source and says its examples and inequality answer the questions posed there. The local run indexes contained no prior packet for arXiv:1111.0460; the decisive citation chain is arXiv:1207.0946, Theorems 1.1–1.2 and their remarks. No claim is made about source Question 3.

References

[1] E. Hernández, *Lebesgue-Type inequalities for quasi-greedy bases*, arXiv:1111.0460 (2011).

[2] G. Garrigós, E. Hernández, and T. Oikhberg, *Lebesgue-Type inequalities for quasi-greedy bases*, arXiv:1207.0946 (2012).